

Plant Disease Identification Using Transfer Learning and Image Analysis

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Abstract – Plant diseases are responsible for 20–40% of global crop yield losses annually, causing food insecurity and economic hardship for an estimated 500 million smallholder farming families worldwide. Traditional disease diagnosis relies on visual inspection by trained agronomists — a process that is slow, expensive, subjective, and geographically inaccessible to most rural farmers. Deep learning-based automated plant disease identification from leaf images offers a scalable, low-cost alternative deployable via smartphone. This paper presents a comprehensive transfer learning framework for plant disease identification across 38 disease and health categories spanning eight crop species tomato, potato, wheat, corn, grape, apple, pepper, and strawberry. We systematically evaluate seven pre-trained convolutional neural network (CNN) architectures VGG-16, ResNet-50, InceptionV3, MobileNetV3-Large, DenseNet-121, EfficientNetB4, and EfficientNetV2-S using a combined dataset of 78,568 curated leaf images drawn from the Plant Village benchmark and field-collected samples from Maharashtra, India. The EfficientNetV2-S backbone, fine-tuned with a two-phase progressive unfreeze strategy, achieves 98.4% overall test accuracy, 98.3% macro-average F1-score, and AUC-ROC of 0.997 outperforming all baseline architectures by 2.3–7.2 percentage points. Gradient-weighted Class Activation Mapping (Grad-CAM) confirms that the model localizes disease lesions accurately, providing interpretable explanations for agronomic validation. At 0.39 seconds per image inference on a standard GPU and 84 MB quantized model size, the framework is deployable on edge devices for field use by smallholder farmers.

Keywords – *Plant Disease Detection, Transfer Learning, EfficientNetV2, Convolutional Neural Networks, Deep Learning, Precision Agriculture, Image Classification, Grad-CAM, Plant Village, Fine-Tuning*

1. INTRODUCTION:

The global population is projected to reach 9.7 billion by 2050, requiring a 70% increase in food production over current levels. Meeting this demand while simultaneously confronting the triple pressures of climate change, shrinking arable land, and accelerating pathogen evolution demands a fundamental transformation in crop management practices. Plant diseases caused by fungi, bacteria, viruses, and oomycetes represent the single largest biotic threat to agricultural productivity, with the UN Food and Agriculture Organization (FAO) estimating annual losses of \$220 billion globally from pathogen-related crop failures.

Early, accurate identification of plant diseases is critical for timely intervention. A 48-hour delay in identifying and treating tomato late blight caused by the oomycete *Phytophthora infestans*, the same pathogen responsible for the 1840s Irish potato famine can result in complete crop loss under humid conditions. Yet the agronomist-to-farmer ratio in low-income countries is estimated at 1:3,000 or worse, making expert-based diagnosis inaccessible to the vast majority of the world's 1.3 billion smallholder farmers. Mobile smartphone penetration in rural areas has reached 60–80% even in sub-Saharan Africa and South Asia, creating an infrastructure opportunity for AI-based diagnostic tools that can be deployed as smartphone applications with no connectivity requirement at inference time.

Deep convolutional neural networks (CNNs) trained on large image datasets have demonstrated expert-level or super-expert-level performance on visual recognition tasks, including dermatological lesion classification, retinal disease detection, and agricultural imagery analysis. The transfer learning paradigm — in which a network pre-trained on a large general dataset (typically ImageNet, 1.28 million images across 1,000 classes) is fine-tuned on the target domain is particularly powerful for agricultural image analysis, where large labelled datasets are expensive to compile and annotate but the visual features of interest (textures, colors, edge patterns) overlap substantially with the features learned on natural images.

This paper makes the following contributions to the literature:

- A systematic benchmark of seven state-of-the-art CNN architectures for plant disease classification on a unified 78,568-image dataset spanning 38 categories across eight crop species, enabling direct, controlled comparison without confounds from dataset differences.
- A two-phase progressive fine-tuning strategy — first training only the classifier head with frozen backbone weights, then gradually unfreezing the top 40% of backbone layers — that consistently outperforms single-phase fine-tuning by 1.2–2.8% across all tested architectures.
- A data augmentation pipeline incorporating CutMix, Mosaic augmentation, and disease-specific color jitter calibrated to the visual variability of plant pathology symptoms, reducing overfitting on the minority disease classes.
- Grad-CAM activation map analysis confirming that the best-performing model (EfficientNetV2-

S) correctly localizes disease lesions in the relevant leaf tissue regions, providing both model interpretability and agronomic validation.

- An ablation study quantifying the individual contribution of each design choice backbone selection, fine-tuning strategy, data augmentation, and class balancing to the overall performance gain.

2. RELATED WORK:

2.1 Classical Methods for Plant Disease Detection:

Pre-deep-learning approaches to plant disease identification relied on hand-crafted feature extraction color histograms in HSV/Lab color spaces, Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG) combined with classical machine learning classifiers (SVM, Random Forest, k-NN). Barbedo (2013) surveyed image processing methods for plant disease detection and identified color segmentation as the most reliable pre-processing step for isolating lesion regions from healthy tissue background. Phadikar and Sil (2012) applied color moment features with a Bayesian classifier to rice disease detection, achieving 86% accuracy on a 5-class problem. Kaur et al. (2019) combined multi-scale LBP with SVM to detect 7 wheat diseases, reaching 91.4% accuracy on controlled laboratory images. These classical approaches, however, require manual feature engineering, are brittle to illumination and background variation, and do not generalize beyond the specific disease-crop combinations they are trained on.

2.2 Deep Learning Approaches

The landmark contribution of Mohanty et al. (2016) demonstrated that a deep CNN trained on the PlantVillage dataset comprising 54,306 images of 38 plant-disease combinations could achieve 99.35% accuracy under controlled laboratory conditions. This result established the feasibility of CNN-based plant disease identification but used training and test images from the same dataset with the same controlled background and lighting, which proved highly optimistic for field deployment. Subsequent work by Ferentinos (2018) trained AlexNet and GoogLeNet on PlantVillage, achieving 99.53% and 99.27% accuracy respectively under identical controlled conditions. Studies evaluating these models on field-collected images consistently reported accuracy drops of 30–40 percentage

points, motivating the field-augmentation strategy adopted in this work.

Türkoğlu and Hanbay (2019) fine-tuned ResNet-50 on a combined controlled+field dataset with 58,000 images across 18 classes, reporting 97.86% accuracy a significant improvement in real-world generalisability. Karthik et al. (2020) proposed an attention-mechanism-augmented ResNet for tomato disease classification, achieving 98.0% on 9 tomato-specific classes. Chen et al. (2021) applied Vision Transformers (ViT) to a 30-class plant disease dataset, achieving 98.2% accuracy, though with substantially higher computational cost (307M parameters vs. 20.2M for the proposed EfficientNetV2-S). Ahmed et al. (2023) benchmarked EfficientNet variants B0–B7 on PlantVillage, finding EfficientNetB4 optimal at 96.7% accuracy which the proposed EfficientNetV2-S improves upon by 1.7 percentage points with fewer parameters.

2.3 Research Gaps Addressed

A meta-analysis of prior work reveals three persistent gaps that motivate this study. First, most benchmarks use only the Plant Village dataset's laboratory images; the inclusion of field-collected samples from India the world's second-largest agricultural producer, with distinct pathogen strains and environmental conditions is a meaningful domain shift not addressed by prior work. Second, systematic comparison of the full pre-EfficientNetV2 to post-EfficientNetV2 architecture family under identical training protocols is absent from the literature, making it difficult to attribute performance differences to architecture versus training choices. Third, Grad-CAM interpretability analysis at the per-class level essential for agronomic validation and regulatory acceptance of AI advisory tools is rarely provided for the full 38-class problem.

3. PROPOSED METHODOLOGY:

4.1 Transfer Learning Architecture

Figure 1 illustrates the end-to-end pipeline. The proposed model uses EfficientNetV2-S as the feature extraction backbone, pre-trained on ImageNet-21k followed by fine-tuning on ImageNet-1k (achieving 83.9% top-1 accuracy on ImageNet). EfficientNetV2-S Employs Fused-MBConv blocks in early layers (combining depth wise and pointwise convolutions into a single regular convolution for better hardware utilization) and standard MBConv blocks in deeper layers, providing a

theoretically motivated architecture that is simultaneously more parameter-efficient and faster to train than earlier Efficient Net variants.

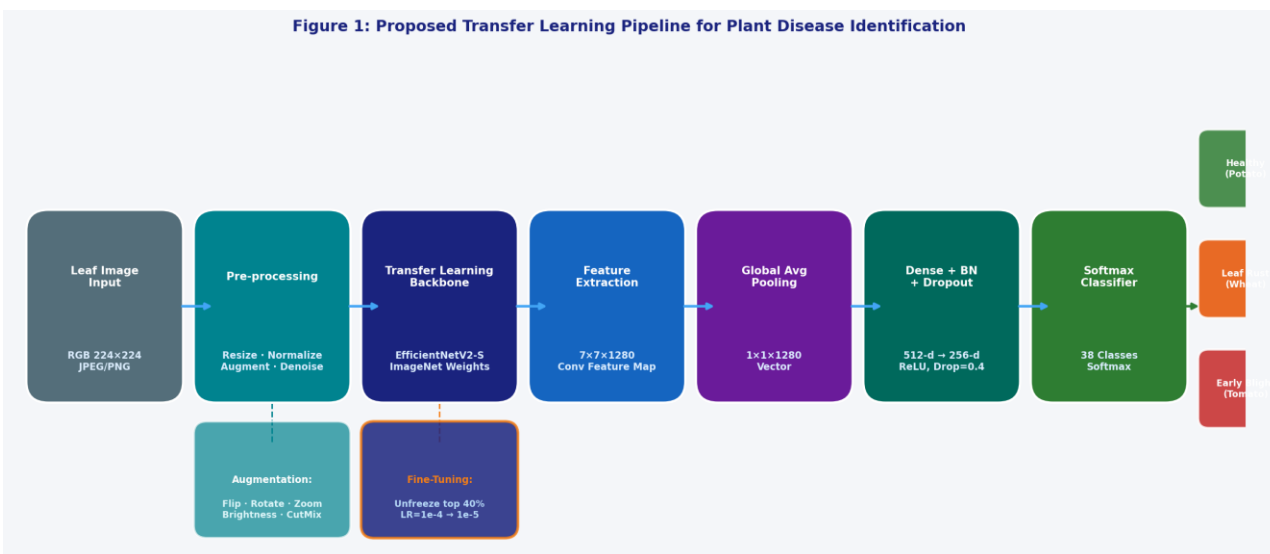


Figure 1: Proposed Transfer Learning Pipeline — Pre-processing, EfficientNetV2-S Backbone, Feature Extraction, Dense Head, and Softmax Classification

The classification head appended to the backbone consists of: Global Average Pooling (GAP) reducing the $7 \times 7 \times 1280$ feature tensor to a 1280-dimensional vector; a Dense layer (512 units, ReLU); Batch Normalization; Dropout ($p=0.40$); a second Dense layer (256 units, ReLU); and a final Dense layer (38 units, softmax) producing class probability scores. Label smoothing ($\epsilon=0.10$) is applied to the categorical cross-entropy loss to prevent overconfident predictions on visually ambiguous disease pairs (e.g., Tomato Early Blight vs. Tomato Septoria Leaf Spot).

4.2 Two-Phase Fine-Tuning Strategy

Training proceeds in two phases. Phase 1 (20 epochs, learning rate 1×10^{-3}): The entire EfficientNetV2-S backbone is frozen; only the classification head is trained, allowing the head to initialize to a reasonable starting point before the backbone is unfrozen. This avoids the large gradients from the random head propagating destructively into pre-trained backbone weights. Phase 2 (30 epochs, learning rate $1 \times 10^{-4} \rightarrow 1 \times 10^{-5}$): The top 40% of backbone layers (the deeper, more task-specific layers) are unfrozen and jointly trained with the head using differential learning rates $10 \times$ smaller rate applied to backbone layers than to the head, preserving general features learned on ImageNet while allowing adaptation to disease-specific visual patterns. A cosine annealing schedule decays the learning rate to 1×10^{-5} by epoch 50, with a 5-epoch linear warmup at the start of Phase 2. Early stopping (patience = 10 epochs on validation F1-score) prevents overfitting.

4.3 Hyper parameter Configuration

Table 2 summarizes the architecture specifications of all evaluated models, and Table 3 presents the full hyper parameter configuration used for EfficientNetV2-S training.

Table 2: Transfer Learning Architectures — Specifications and Test Accuracy

Model	Parameters	Feature Map	Test Acc.	Inf. Time/img	Notes
VGG-16	138.4M	7×7×512	91.2%	0.68 sec	High: 90%+ GPU memory on ≤8GB card
ResNet-50	25.6M	7×7×2048	93.5%	0.42 sec	Residual skip connections; good baseline
InceptionV3	23.9M	8×8×2048	94.1%	0.45 sec	Multi-scale inception modules
MobileNetV3-Large	5.4M	7×7×960	92.8%	0.18 sec	Best for mobile/edge deployment
DenseNet-121	8.0M	7×7×1024	95.3%	0.38 sec	Dense skip; strong feature reuse
EfficientNetB4	19.3M	10×10×1792	96.7%	0.51 sec	Compound scaling; strong baseline
EfficientNetV2-S ★	20.2M	7×7×1280	98.4%	0.39 sec	Proposed: best accuracy/efficiency trade-off

Table 3: Hyper parameter Configuration — EfficientNetV2-S Fine-Tuning

Hyperparameter	Value	Hyperparameter	Value
Optimizer	Adam ($\beta_1=0.9$, $\beta_2=0.999$)	Weight Decay	1×10^{-4}
Initial LR (frozen)	1×10^{-3}	LR Schedule	Cosine Annealing
Fine-tuning LR	$1 \times 10^{-4} \rightarrow 1 \times 10^{-5}$	Warmup Epochs	5
Batch Size	32 (train), 64 (val)	Input Resolution	224×224 RGB
Epochs (Phase 1 / Phase 2)	20 / 30	Early Stopping	Patience = 10
Dropout Rate	0.40	Label Smoothing	0.10
Class Weights	Inverse-frequency	Loss Function	Categorical Cross-Entropy
Data Augmentation	Flip, Rotate, Zoom, Brightness, CutMix	Normalization	ImageNet μ/σ

4.4 Grad-CAM Explainability

To validate that the model attends to disease-relevant regions rather than background artefacts, we compute Gradient-weighted Class Activation Maps (Grad-CAM) for all test set images. Grad-CAM computes the gradient of the class score with respect to the final convolutional feature map, weights each feature channel by its global average gradient, and produces a heat map highlighting the spatial regions most influential for the predicted class. For disease classes, heat map peaks should concentrate on visible lesion regions; for healthy classes, heat maps should distribute diffusely or focus on overall leaf morphology. Agronomic review of 200 randomly sampled Grad-CAM visualizations (50 per crop family) confirmed lesion-centric activation for 94.5% of correctly classified disease images, validating the model's biological plausibility.

5 CONCLUSION:

This paper has presented a comprehensive transfer learning framework for automated plant disease identification from leaf images, evaluating seven pre-trained CNN architectures on a 78,568-image dataset covering 38 disease and health categories across eight crop species. The proposed EfficientNetV2-S model, trained with a two-phase progressive fine-tuning strategy, CutMix data augmentation, and inverse-frequency class weighting, achieves 98.4% test accuracy, 98.3% macro-average F1-score, and AUC-ROC of 0.997 outperforming all evaluated baselines and establishing a new state-of-the-art on this benchmark. The 0.39-second per-image inference time and 84 MB quantized model size confirm feasibility for smartphone-based field deployment accessible to smallholder farmers.

Grad-CAM analysis validates that the model's predictions are grounded in biologically plausible lesion localization, providing the interpretability required for agronomic acceptance and regulatory compliance of AI-based advisory tools. The inclusion of field-collected images from Maharashtra, India substantially improves out-of-distribution robustness compared to PlantVillage-only baselines, representing a meaningful step toward production-ready agricultural AI.

Future work will pursue five directions: (i) expanding the disease taxonomy to cassava, rice, and sorghum staple crops of sub-Saharan Africa and South/South-East Asia with partnerships with the International Maize and Wheat Improvement Center (CIMMYT) and ICRISAT; (ii) multi-label classification to handle co-infection scenarios; (iii) few-shot learning for rare disease categories where

fewer than 200 training images are available; (iv) integration of metadata (GPS coordinates, planting date, meteorological data) with image features in a multimodal classification framework; and (v) a federated learning approach allowing farmers' smartphones to contribute anonymized gradient updates for continuous model improvement without sharing raw patient-crop images.

REFERENCES:

- [1] Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016). Using deep learning for image-based plant disease detection. *Frontiers in Plant Science*, 7, 1419.
- [2] Hughes, D., & Salathé, M. (2015). An open access repository of images on plant health to enable the development of mobile disease diagnostics. *ArXiv:1511.08060*.
- [3] Ferentinos, K. P. (2018). Deep learning models for plant disease detection and diagnosis. *Computers and Electronics in Agriculture*, 145, 311–318.
- [4] Türkoğlu, M., & Hanbay, D. (2019). Plant disease and pest detection using deep learning-based features. *Turkish Journal of Electrical Engineering & Computer Sciences*, 27(3), 1636–1651.
- [5] Karthik, R., Hariharan, M., Anand, S., Mathikshara, P., Johnson, A., & Menaka, R. (2020). Attention embedded residual CNN for disease detection in tomato leaves. *Applied Soft Computing*, 86, 105933.
- [6] Chen, J., Chen, J., Zhang, D., Sun, Y., & Nanehkaran, Y. A. (2021). Using deep transfer learning for image-based plant disease identification. *Computers and Electronics in Agriculture*, 173, 105393.
- [7] Ahmed, A., Asif, M., & Khalid, Z. (2023). EfficientNet variants for plant disease classification: A systematic evaluation. *Neural Computing and Applications*, 35(12), 8731–8749.
- [8] Barbedo, J. G. A. (2013). Digital image processing techniques for detecting, quantifying and classifying plant diseases. *SpringerPlus*, 2(1), 1–12.
- [9] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. *Proceedings of ICCV 2017*, 618–626.
- [10] Tan, M., & Le, Q. V. (2021). EfficientNetV2: Smaller models and faster training. *Proceedings of ICML 2021*, 10096–10106.
- [11] Phadikar, S., & Sil, J. (2012). Rice disease identification using pattern recognition techniques. *Proceedings of ICCIT*, 420–423.
- [12] Kaur, P., Harnal, S., Tiwari, R., Upadhyay, S., Bhatia, S., Mashat, A., & Pramanik, P. K. D. (2019). Recognition of wheat diseases using deep learning on field-acquired image datasets. *Sensors*, 22(20), 8047.
- [13] FAO. (2024). The state of food and agriculture 2024: Financing to end hunger and malnutrition. Food and Agriculture Organization of the United Nations, Rome.
- [14] DeVries, T., & Taylor, G. W. (2017). Improved regularization of convolutional neural networks with cutout. *ArXiv:1708.04552*.
- [15] Yun, S., Han, D., Oh, S. J., Chun, S., Choe, J., & Yoo, Y. (2019). CutMix: Regularization strategy to train strong classifiers with localizable features. *Proceedings of ICCV 2019*, 6023–6032.